



Our optimized airfoil delays aircraft stalling by 61%,



increasing fuel efficiency by ~10%

Research Question

How does the long gap width (2mm, 3mm, 4mm, 5mm) and location (10%, 15%, 20%, 25%) of vortex generators along a NACA 2412 airfoil affect its stall angle in low speeds (3.02 km/h) at low Reynolds numbers?

Abstract

The high cost of flight training creates barriers for aspiring pilots. Fuel is a major factor in these high costs, accounting for ~20% of flight school operating fees.¹⁷ This study improves the performance of the common NACA 2412 training airfoil by using new prospective technologies such as vortex generators (VGs).¹ Implementing VGs on training aircraft will significantly reduce fuel costs, lowering the barrier for aspiring pilots. It also reduces emissions, which leads to an increase in sustainability while supporting the next generation of aviators.

Introduction

- The boundary layer is a layer of slow airflow on the wing's surface and is a key source of drag and stalls when it separates from the wing's surface. (Figure 1)
 - Separation occurs due to adverse pressure gradients (APG) (Figure 2)
- Vortex generators (VGs) are small fins that delay this separation. (Figure 3)
 - VGs create vortices that mix the boundary layer with faster free-stream air.
 - This energizes the boundary layer, allowing it to stay attached to the wing for longer.⁷
 - The wing can thus reach higher angles of attack (AOA) without stalling.
 - Although VGs are popular in commercial airplanes and the automobile industry, their effectiveness in beginner airplanes is less explored.
- Reynolds number: The ratio of inertial to viscous forces in a fluid. It is a measure of how turbulent or laminar a fluid is.
- VGs have eleven performance variables (Figure 3).³
 - Long gap spacing and chordwise location were two variables that we investigated due to a lack of prior research.

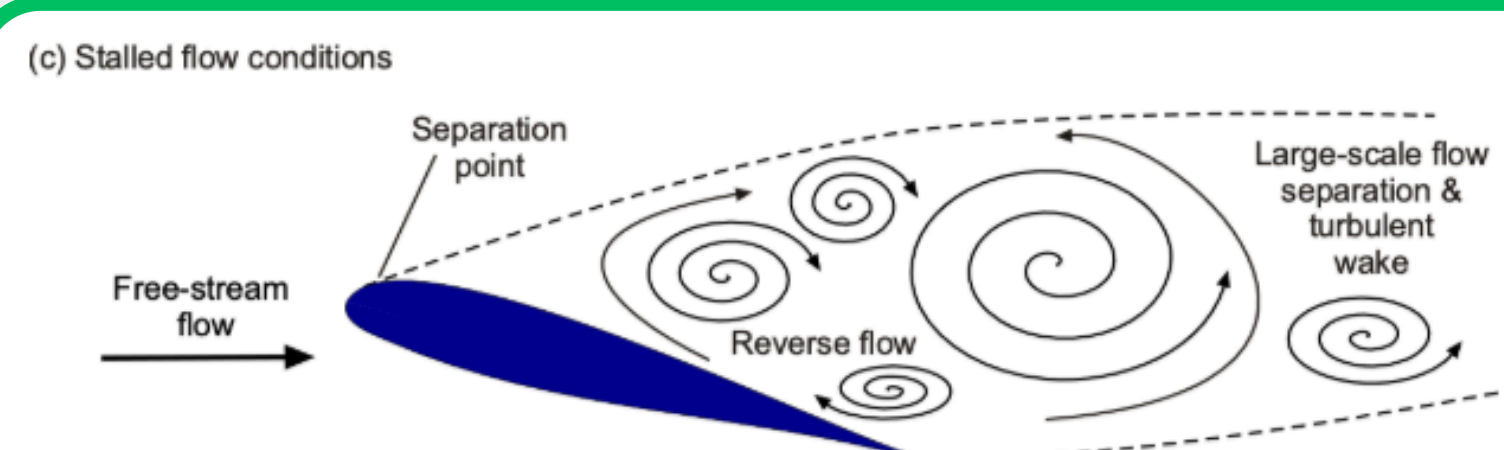


Figure 1. Boundary layer separation

Leishman, J. G. (n.d.). Introduction to boundary layers. In Introduction to aerospace flight vehicles. Eagle Pubs. Retrieved May 18, 2025, from <https://eaglepubs.erau.edu/introductiontoaerospaceflightvehicles/chapter/introduction-to-boundary-layers/>

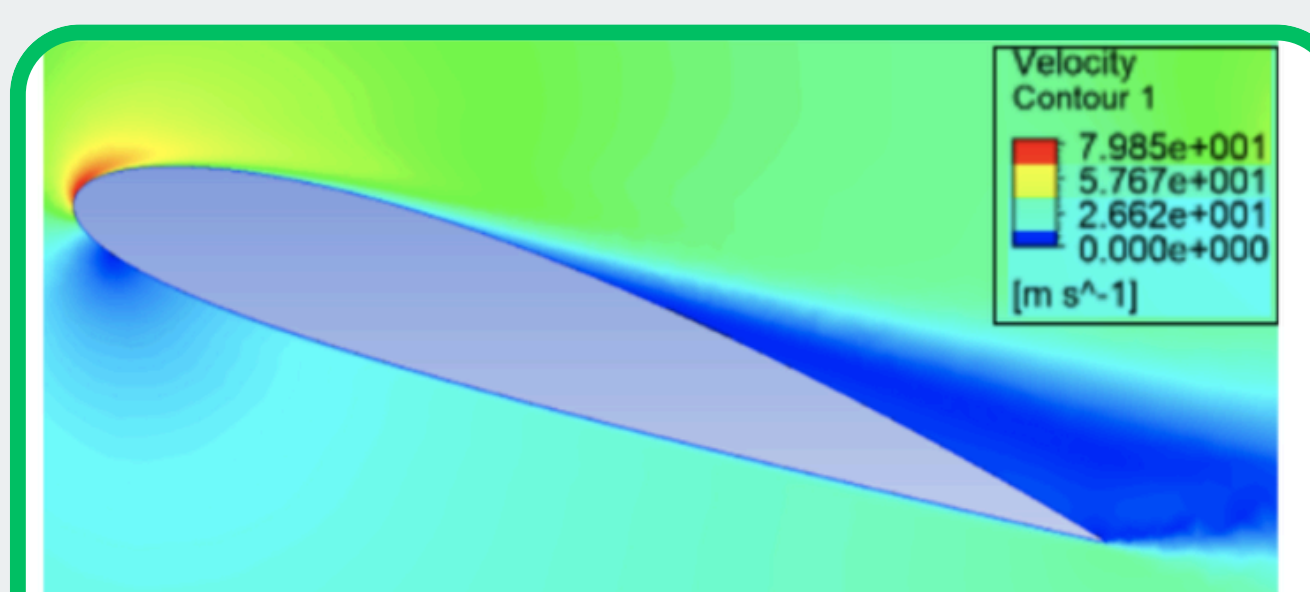


Figure 2. Pressure gradient around an airfoil

Blak, B., & Jindel, T. K. (2021, February 1). Effect of co-flow jet characteristic on the aerodynamic performance of an airfoil. <https://www.jst.org/2021-february/2021-11/effect-of-co-flow-jet-characteristic-on-the-aerodynamic-performance-of-an-airfoil>

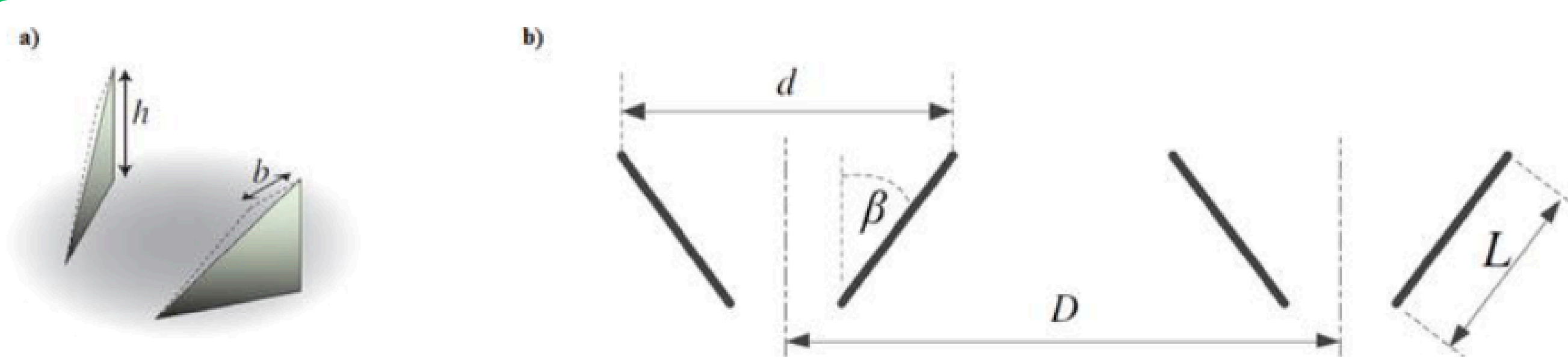


Figure 3. Various variables labeled on a VG.

Genç, M. S.; Koca, K.; Demir, H.; Açıkel, H. H. Traditional and new types of passive flow control techniques to pave the way for high maneuverability and low structural weight for UAVs and MAVs. IntechOpen 2020. <https://www.intechopen.com/chapters/70899>.

Methods

Wind Tunnel

The wind tunnel was constructed manually with these key components (Figure 4):

- An airflow filter to decrease the Reynolds number.
- A suction fan to mimic air flowing around an airfoil.
- A removable back plate to swap airfoils between trials (Figure 5).
- A protractor to measure the Angle of Attack (AOA).
- Smoke to visualize airflow inside the wind tunnel. The wind tunnel had an airspeed of 3.02 km/h.

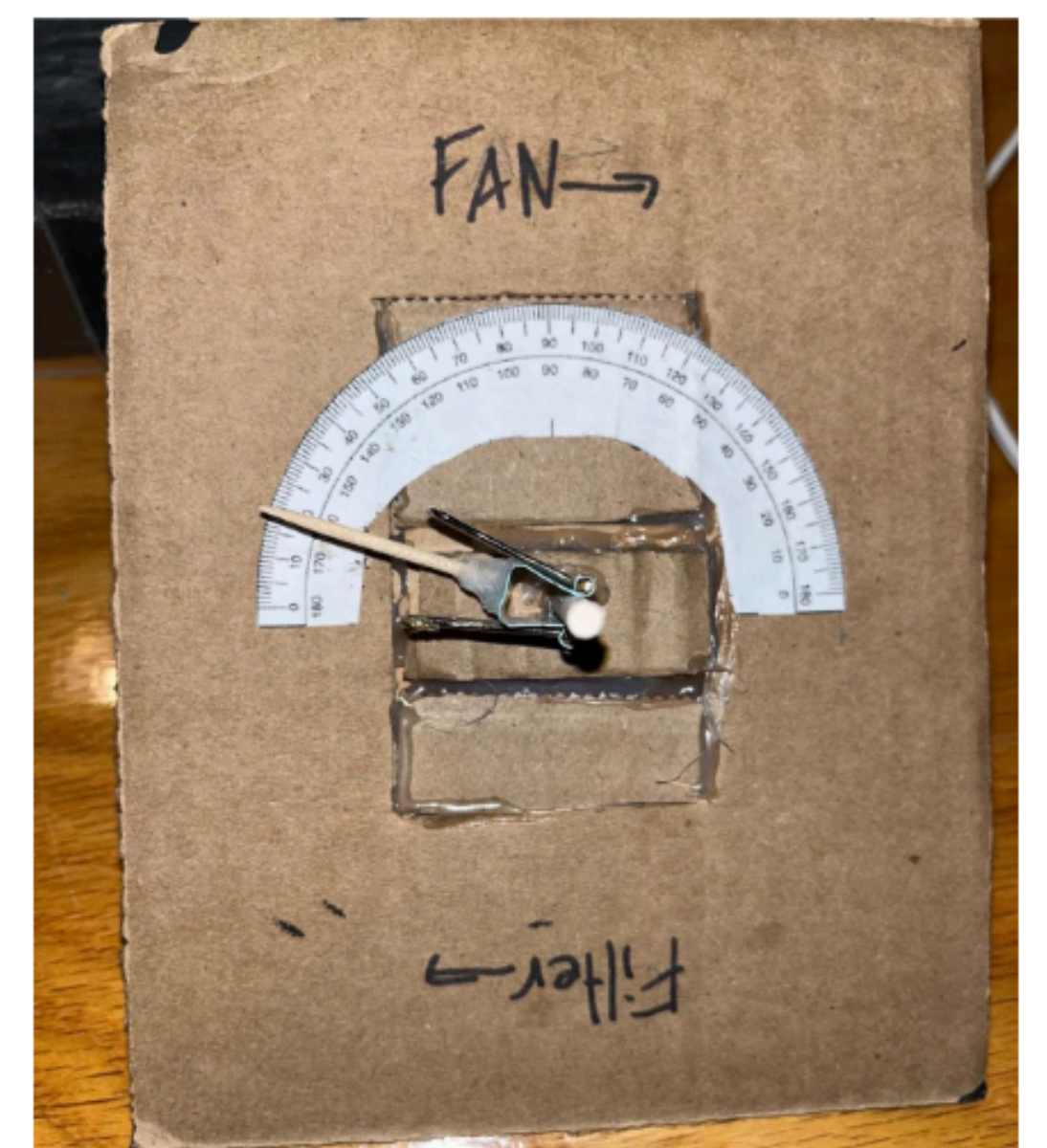


Figure 5: Removable backplate with protractor attached.

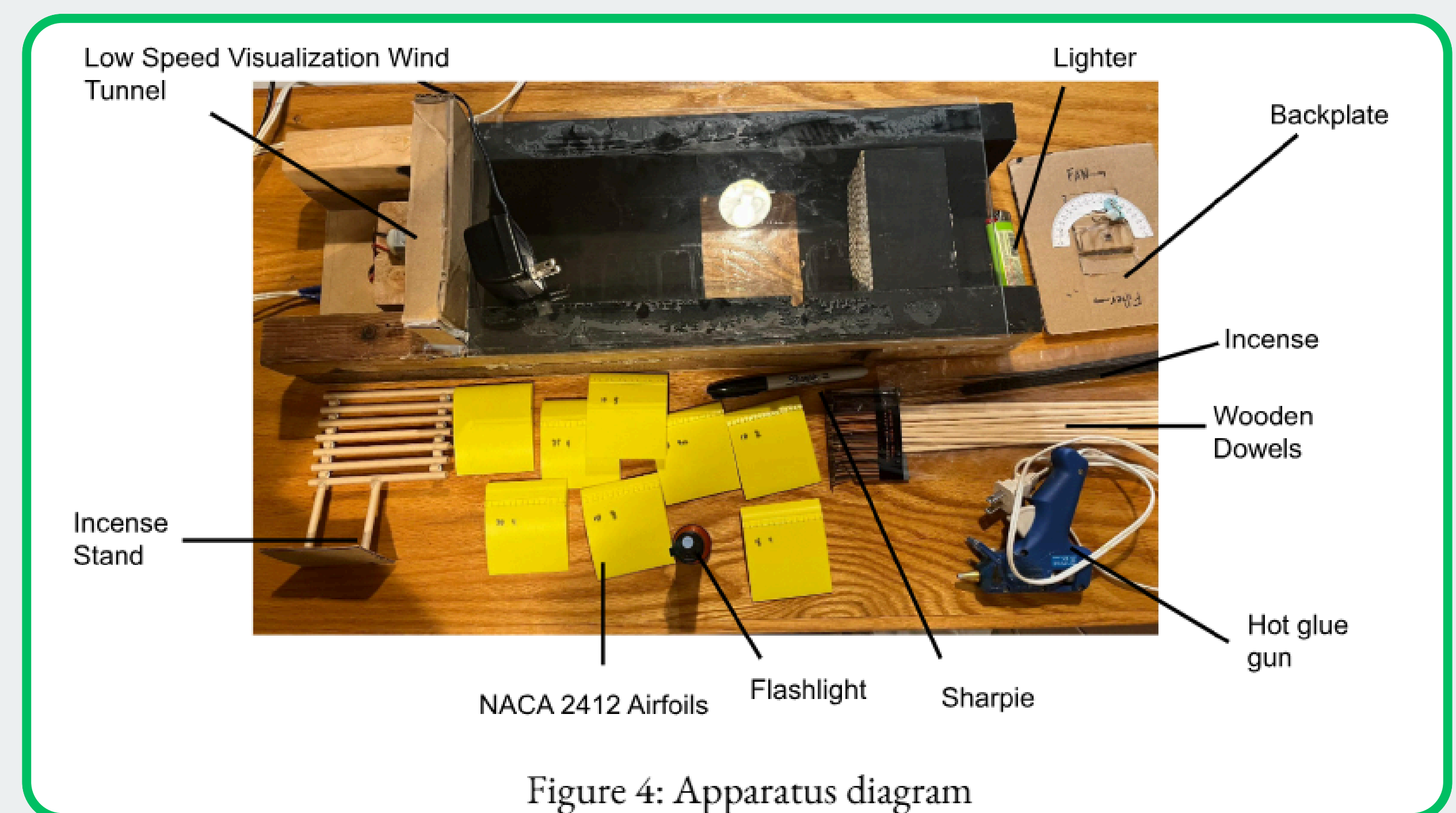


Figure 4: Apparatus diagram

Airfoils

- The NACA 2412 airfoils were mathematically graphed and imported into CAD software¹¹ (Figures 6 & 7).
- Eight airfoils scaled to 75mm by 65mm were printed, including one airfoil without VGs (control).
- Each vortex generator row was separately modelled and attached.

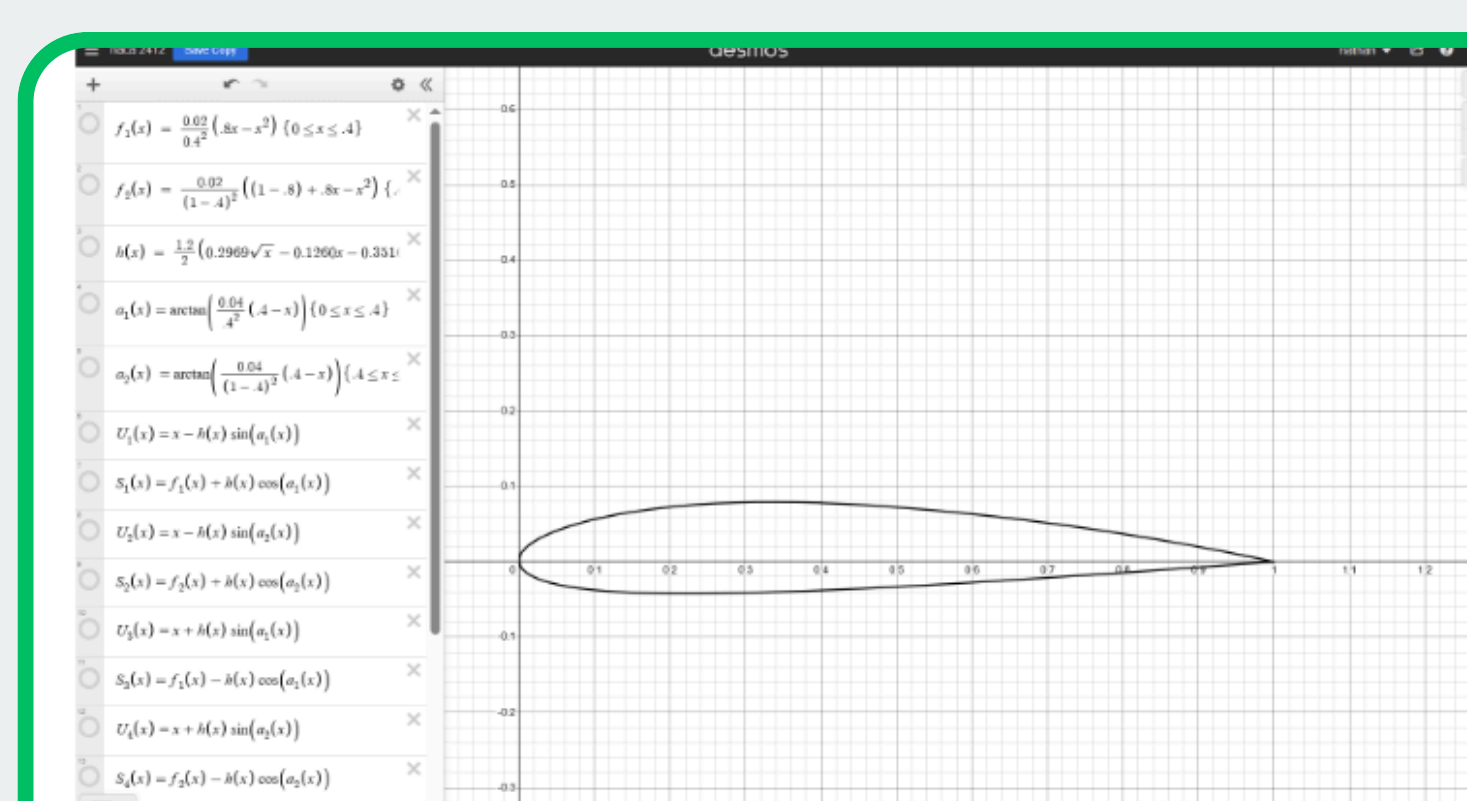


Figure 6. Graphed airfoil on Desmos

Desmos. (2024). Desmos graphing calculator [Online graphing calculator]. <https://www.desmos.com/calculator>

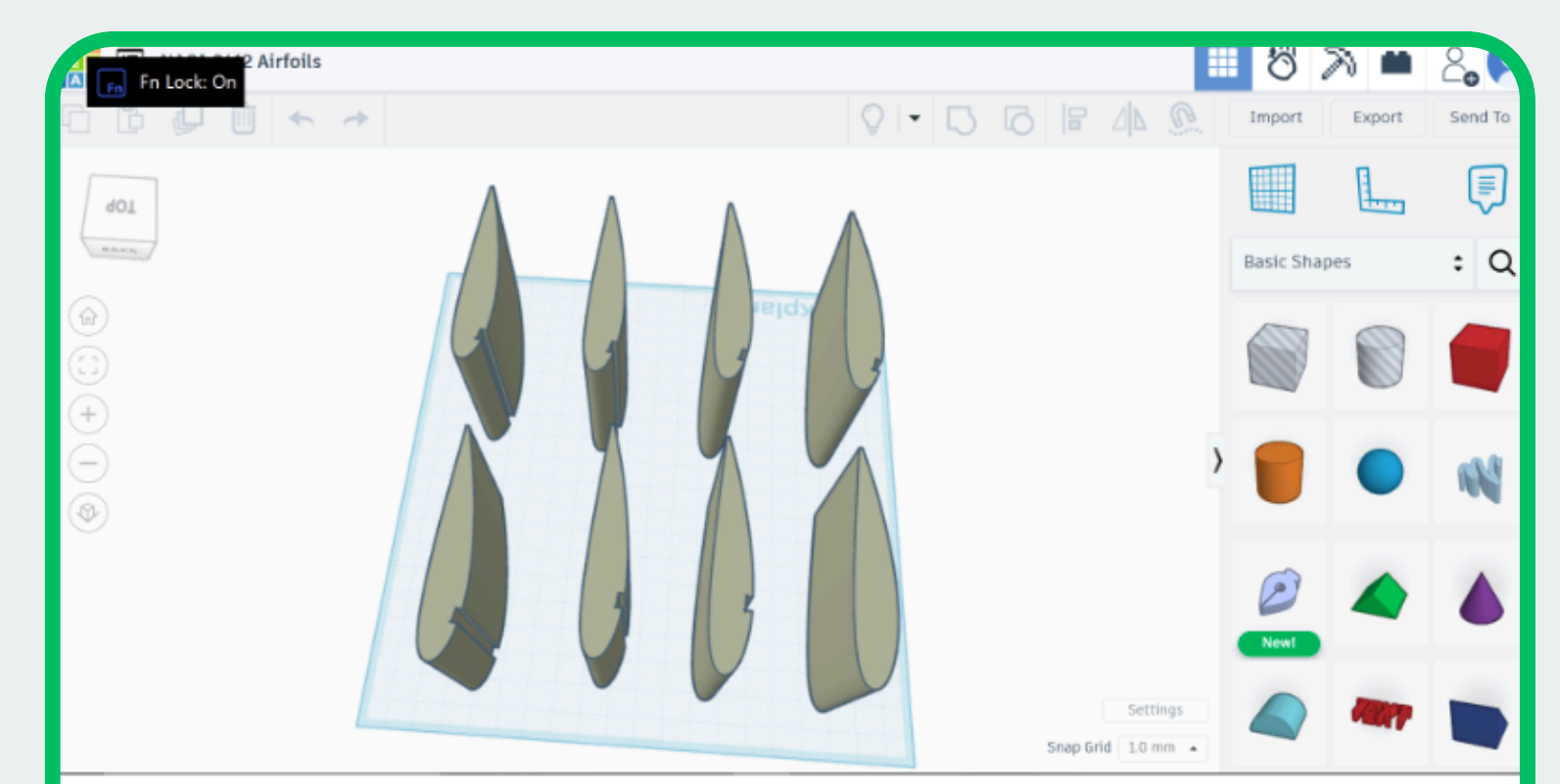


Figure 7. 3D modelled airfoil on Tinkercad

Autodesk. (2024). Tinkercad [3D modeling software]. <https://www.tinkercad.com>

Variables

Controlled:

- VG Height: 2mm¹²
- VG Vane length: 4.38mm (5.8% chord length)¹³
- VG Shape: Triangular¹⁴
- VG Type: Triangular Vane; Counter Rotating¹⁵
- VG Inflow angle: 12°¹⁵
- 3.02 km/h airspeed
- Wind tunnel Reynolds number
- Testing chamber volume
- Boundary layer separation point (30% chord length)
- Airfoil Type: NACA 2412

Manipulated Variables:

- VG Long gap spacing: 2mm, 3mm, 4mm, 5mm)
- VG Location: 10%, 15%, 20%, 25% of the chord length (75mm)

Responding Variable

- Stalling AOA

The manipulated variable values were built off of past research papers. These papers were already published in reputable scientific journals.

Procedure

1. Begin by gathering all materials (Figure 4)
2. Mark out the 30% airfoil length point on each airfoil from the leading edge and glue the wooden dowel.
3. Insert the incense sticks into the stand and ignite them to produce smoke.
4. Insert the airfoil into the wind tunnel and turn on the wind tunnel. Ensure that the airfoil is initially at 0° AOA.
5. Using the flashlight to increase smoke visibility, slowly increase the AOA until the airflow is separated from the airfoil at the marked point. (Figure 8)
6. Record the AOA and reset the airfoil to 0°
7. Repeat steps 4-6 for 3 trials.
8. Repeat for all 8 airfoils.

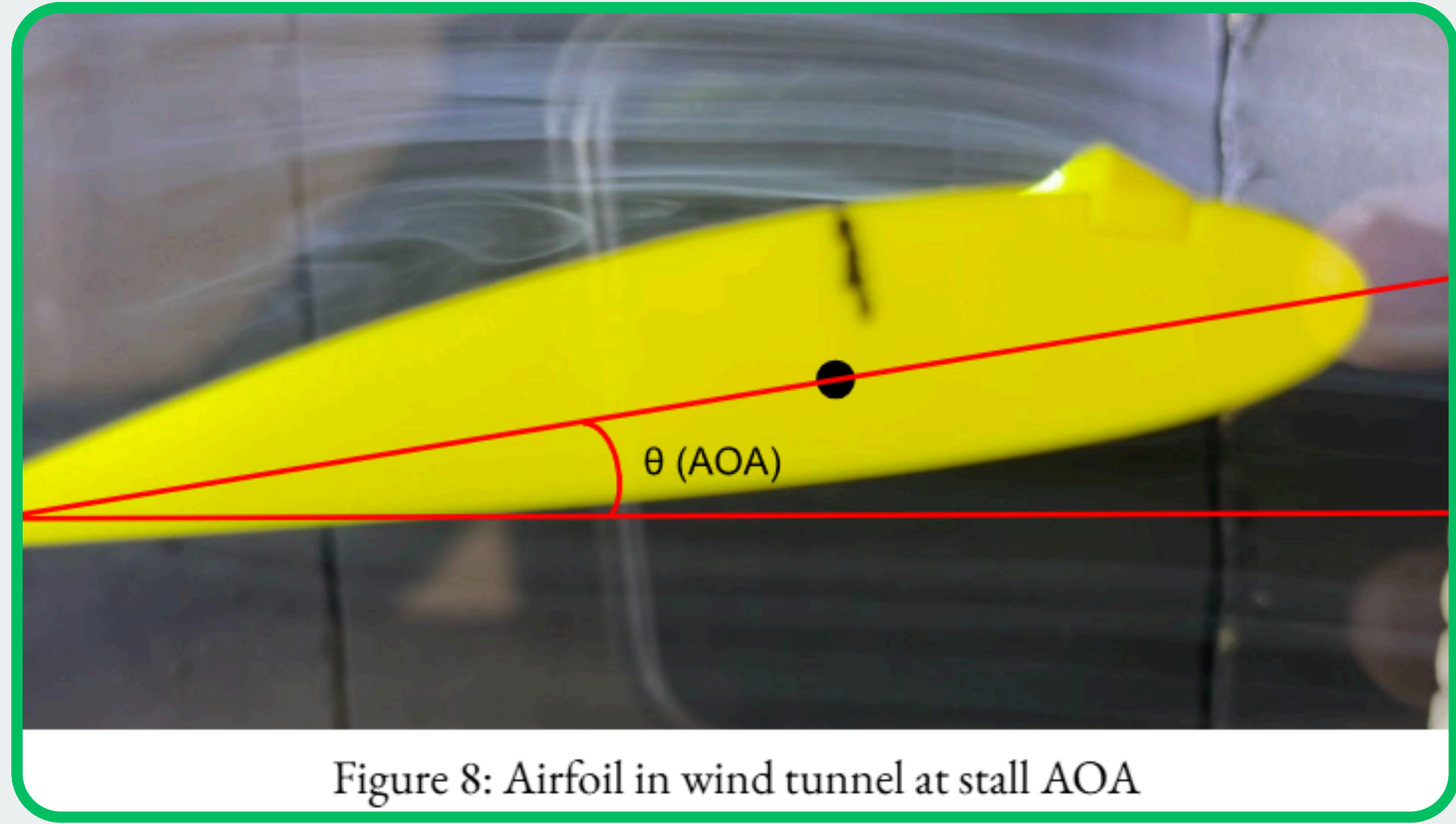


Figure 8: Airfoil in wind tunnel at stall AOA

Stalling usually occurs when the separation point reaches the point of max thickness, which is located at 30% chord length on a NACA 2412.¹⁶ Furthermore, vortex generators mitigate boundary layer separation in general, and therefore choosing the separation point to record the AOA is arbitrary.

Data

The average of the three trials for location produces the values in Table 4, which summarizes the AOA for different VG locations, given a fixed long gap spacing of 4mm. The highest AOA of 11.5° was measured at a placement of 20% of the chord, and the lowest value, 6.8°, occurred at 25%. The average of the three trials recorded for long gap spacing is explored in Table 5, with a fixed location of 10% chord length. 2mm, the smallest gap spacing, resulted in the lowest average AOA of 5.2°, whereas a gap spacing of 4mm produced the highest average AOA of 7.7°.

For the location and long gap spacing experiments, three trials were run for each airfoil, giving a sample size of n = 3. The sample size was determined through scientific discretion, considering that the time required to run the experiment would become unreasonable if the sample size increased further. It is important to note that although 7.7° is higher than the AOA of the control (6.7°), it is significantly lower than the maximum AOA recorded from optimizing location (11.5°). This indicates that placement along the chord is more critical than gap spacing.

Table 4. Average AOA for boundary layer separation at various locations

Airfoil Type	Average AOA (Degrees)
Control	6.7
10% @ 4mm	7.7
15% @ 4mm	11.2
20% @ 4mm	11.5
25% @ 4mm	6.8

Table 5. Average AOA for boundary layer separation at various long gap spacings

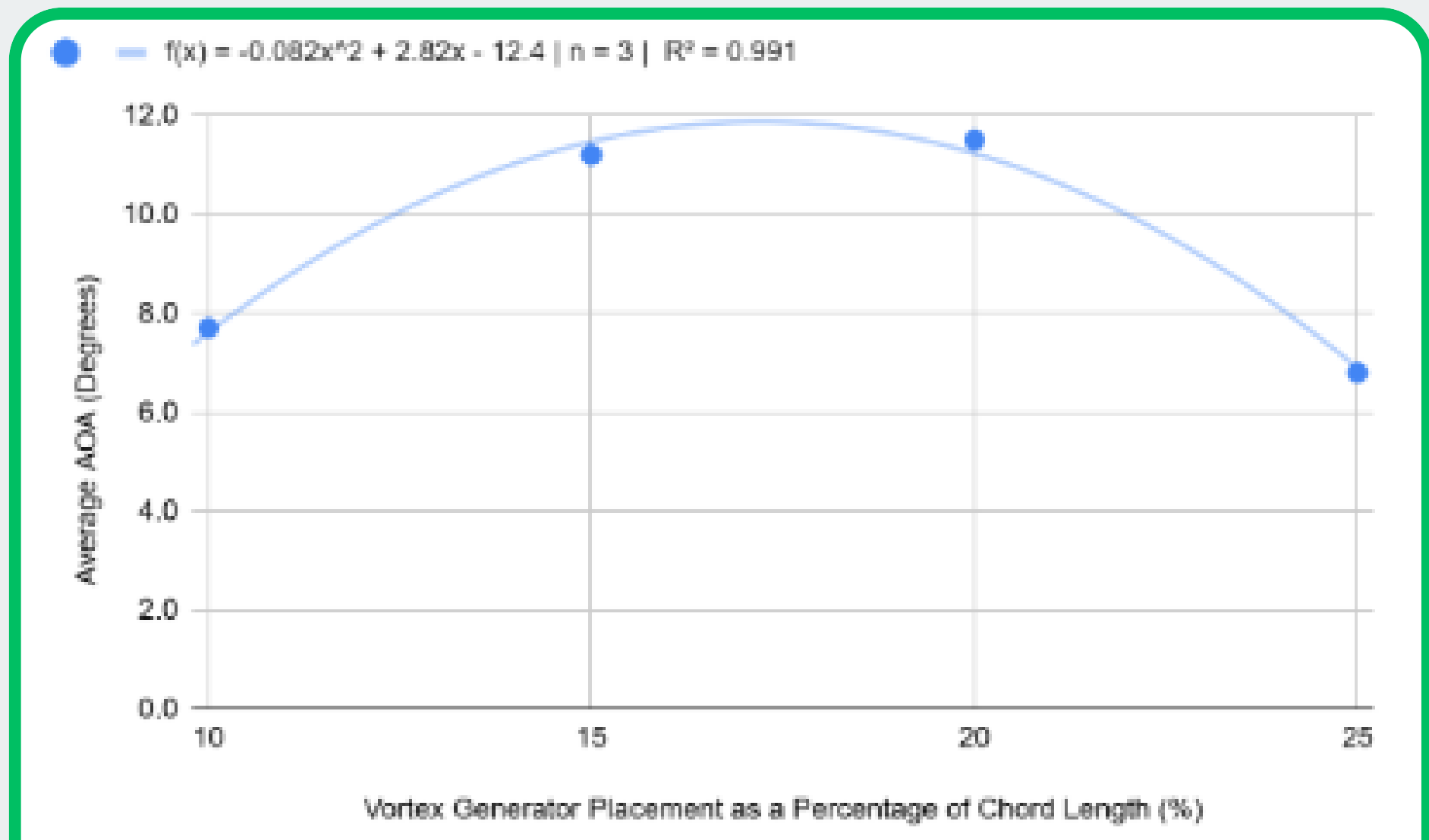
Airfoil Type	Average AOA (Degrees)
10% @ 2mm	5.2
10% @ 3mm	6.8
10% @ 4mm	7.7
10% @ 5mm	5.7

Analysis

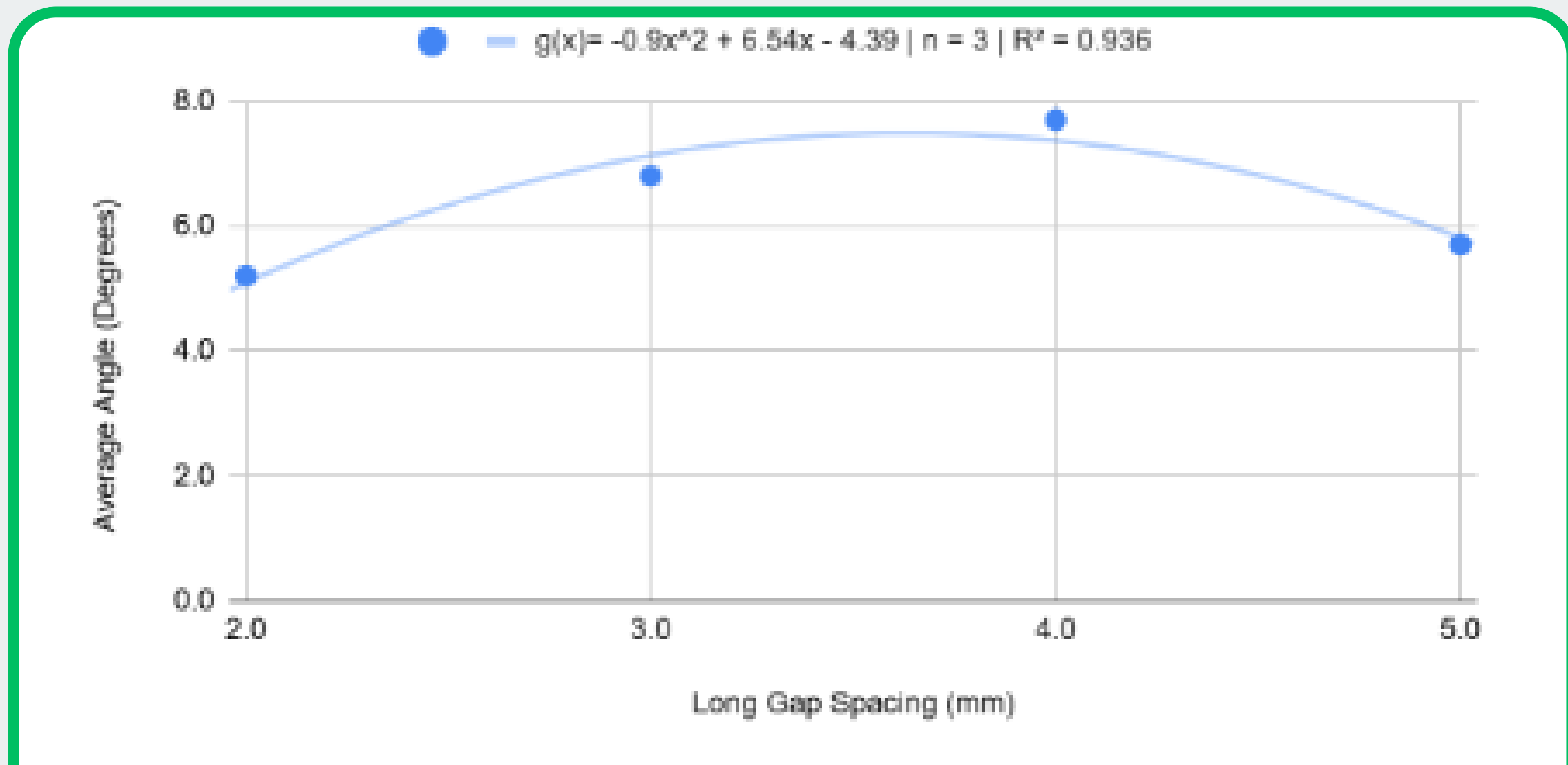
The values in Tables 4 and 5 were plotted on Graphs 1 and 2, respectively. A quadratic trendline was used to model the data in both graphs, which produced an R² value of 0.991 and 0.936, respectively. Since no definitive equation exists that explains the relationship between stall angle and VG parameters, a quadratic trendline was chosen as it best approximates the data and the theory. VG performance increases when approaching the ideal value and then decreases after this value is passed, which produces a clear parabolic relationship. A quadratic was used instead of a higher-order polynomial to reduce the chance of “overfitting” the data. The high R² values in both graphs show a strong correlation between the trendline and the data. This not only justifies the choice to use a quadratic, but it also justifies the choice of having n = 3 since there is negligible random error.

The theoretical best values for the location and long gap spacing are found at the maxima on the trendlines. This means that a VG at 17.2% chord length with a long gap spacing of 3.6mm should have a max AOA of 11.8°. A separate airfoil was 3D printed with these values to compare the theoretical maximum AOA to the experimental maximum AOA. To reduce random error, this airfoil was tested until three trials (n = 3) were recorded within 0.5° of each other (Table 6), which represents the uncertainty of the protractor. N = 3 was once again chosen to match the previous data points.

Averaging the 3 optimized trials (Table 6) produced a final maximum AOA of 10.8°, a 61% increase from the control (6.7°). A 50% delay in flow separation is predicted to increase fuel efficiency by 8%.³ Therefore, a 61% delay should increase fuel efficiency by ~10%. Taking the average fuel cost of flight schools and multiplying the cost by 0.08 shows that \$6000 can be saved on fuel costs per year.



Graph 1. Average boundary layer separation angle as a function of VG location



Graph 2. Average boundary layer separation angle as a function of VG long gap spacing.

Discussion

Placement at 20% of the chord yielded the highest delay in flow separation, with separation occurring at 11.5° AOA, while placement at 25% was least effective, with separation at just 6.8° AOA. This suggests that positioning VGs too far downstream reduces their ability to enhance boundary layer-free stream interaction. The same can be said about positioning the VGs too early, as seen with the 10% placement (7.7°).

A long gap spacing of 2mm (5.2°) and 5mm (5.7°) both produced an AOA that was significantly lower than the AOA of the airfoil without VGs (6.7°). Adding VGs increases the amount of drag created due to form resistance, which also increases the tendency for airflow to separate. Normally, this effect is outweighed by the positive effects of properly optimized VGs. Moreover, the vortices produced by the 2mm VG are likely to destructively interfere with the flow, resulting in less mixing overall. Similarly, the 5mm VG likely produced vortices that were too far apart, resulting in less mixing.

Conclusions

Our optimized airfoil had a 61% AOA increase from the control (6.7°). This is estimated to reduce fuel costs by ~10%³, roughly equating to \$6K saved per year¹⁷ on flight school.

Although the potential for VG implementation on beginner aircraft has been effectively demonstrated, improvements could be made to reduce random error and standard deviation.

1. Increase the number of trials or repeat the experiment until >3 values are obtained within 0.5°.
2. Modify the apparatus to incorporate a more precise protractor

Focusing on making the data scalable to fit real-world planes at higher Reynolds numbers would be the next step in this project. Another area of focus could be optimizing the other 9 VG variables and measuring lift and drag forces.

Table 6. Chosen trials for the optimized airfoil.

Trial #	AOA (Degrees)
3	11
4	11
6	10.5
Average	10.8

Thanks and References

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